INTERIM FEDERAL SPECIFICATION
TRUCKS, LIGHT COMMERCIAL, ELECTRIC POWERED, FRONT OR REAR WHEEL DRIVEN (4x2, 500 TO 3000 POUNDS MINIMUM PAYLOAD)

This Interim Federal Specification dated September 1, 1977, was developed by the General Services Administration, Federal Supply Service, National Automotive Center, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

Please note: When selecting an electric vehicle, extremely careful consideration must be given to the vehicle's performance capabilities as compared to the intended application. Ambient temperature conditions, gradeability requirements, acceleration, operating speeds, range, and overall route profile must be carefully reviewed in order to ensure that an electric vehicle is not placed in an unsuitable application.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers new commercially produced electrically propelled, four-wheeled, two front or rear wheel driven (4x2) light trucks, in the 1/4 to 1-1/2 ton payload range. All vehicles covered by this specification must meet all applicable Federal Motor Vehicle Safety Standards (FMVSS) and Federal Motor Carrier Safety Regulations (FMCSR). Vehicles procured under this specification shall be warranted by the manufacturer or contractor as specified in Section 6.4.

1.1.1 Application. This specification shall be used for the purchase of electric vehicles to be used by Federal agencies. These vehicles can be used in lieu of conventional gasoline or diesel powered vehicles in specialized utilitarian applications (see 6.1). Due to the limited performance, range, and capacity of electric vehicles, the intended use of these vehicles range from in-plant-on-facility administrative use at low speeds for transport of mail, small parcels, maintenance and administrative personnel to on-road transport of larger quantities of mail, cargo, supplies, equipment, personnel, and routine security patrols.

1.1.2 Requisitioning. Requisitions for these vehicle(s) and their equipment should be submitted in accordance with the ordering data (see 6.2) of this specification. These requisitions shall be accompanied by a description setting forth the intended use of the vehicle, and the operating conditions as mentioned in the boxed cautionary note above. These shall be submitted to the General Services Administration, National Automotive Center (FY), Washington, DC 20406.

1.2 Classification. The classification of vehicle(s) shall be by type and class. The dimensional characteristics and the basic minimum requirements for each type are shown in Table I (page 4). Vehicle "types" are determined by the body configurations, and the vehicle class is determined by payload.

1.2.1 Class. The class of vehicle shall be determined by the minimum payload capacity (see 3.6.2 Payload) required for each type of vehicle specified as follows.

Class	Minimum payload (pounds)
A	500
B	1,000
C	1,500
D	2,000
E	3,000

1.2.2 Standard trucks. The vehicles complying with the classified method (see 1.2 and Table I) of this specification are defined as standard electric vehicles for purposes of Government procurement.

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TRUCKS, LIGHT COMMERCIAL, ELECTRIC POWERED FRONT OR REAR WHEEL DRIVEN
(4x2, 500 to 3000 POUNDS MINIMUM PAYLOAD)
SPECIFICATION SUMMARY

Requirement \ Type/Class (Table I, Para. 1.2)			IA	IIA	IIIB	IIIC	IIID	IIIE
Payload - pounds			500	500	1000	1500	2000	3000
Vehicle Type			utility on facility	utility on road	delivery van - on road			
Load Space - cubic feet (Tables III, IV)			25	50	150	225	350	500
Performance based on SAE Elec. Vehicle Test Proced. J227a (Ref. Para. 3.7 & Tab. II)	Range at constant speed - miles	20 MPH	60 miles		-			
		25 MPH	55 "		75 miles			
		30 MPH	45 "		70 "			
		35 MPH	-		60 "			
		40 MPH	-		50 "			
		45 MPH	-		35 "			
	Selected driving patterns - cycles	A	1500 cycles		-			
		B	225 "		250 cycles			
		C	75 "		100 "			
		D	-		45 "			
	Accel. - seconds	0-30 MPH	15 seconds		10 seconds			
		0-40 MPH	-		17 "			
	Gradeability at speed, on 3% grade - MPH		25 MPH		40 MPH			
	Gradeability limit - percent grade		25 percent		30 percent			
Battery Charger (Para. 3.4.3)	Recharge time - hours		Onboard 10 hrs @ 230V line 18 hrs @ 115V line		Offboard 8 hrs @ any specified voltage			
Warranty (Para. 6.4)	Electrical System: Motor, Controller, Wiring, etc. (Para. 6.4.1.1)		3 years/36,000 miles, whichever comes first					
	Propulsion Battery (Para. 6.4.1.2)		3 years/36,000 miles/ 750 charge-discharge cycles, whichever comes first					
	12V Accessory Battery (Para. 6.4.1.3)		3 years					
	Balance of Vehicle (Para. 6.4.1.4)		2 years/24,000 miles, whichever comes first					

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

Fed. Std. No. 297 - Rustproofing of Automotive Vehicles.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Document-, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, Seattle, WA, Philadelphia, PA, and Houston, TX.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-1223 - Administrative Wheeled Vehicles Treatment, Painting, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.

(Copies of Military Specificcations and Standards required by suppliers in connection, with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

49 CFR 393:
Federal Motor Carrier Safety Regulations (FMCSR)
National Highway Traffic Safety Administration (NHTSA)

49 CFR 571:
Federal Motor Vehicle Safety Standards (FMVSS)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

The Tire and Rim Association, Inc. Yearbook:
(Application for copies should be addressed to The Tire and Rim Association, Inc., 3200 West Market Stree, Akron, OH 44313.)

Society of Automotive Engineers (SAF), Inc. Standards and Recommenced Practices:

SAE J227a - Electric Vehicle Test Procedure
SAE J551 - Measurement of Electromagnetic Radiation from a Motor Vehicle
SAE J552 - External Electromagnetic Radiation Suppressors
SAE J678 - Speedometers
SAE J682 - Rear Wheel Splash and Store Throw Protection
SAE J683a - Tire Chain Clearance
SAE J689a - Approach, Departure, and Ramp Breakover Angles
SAE J1024 - Fuel Fired Heaters

(Application for copies should be addressed to the Soceity of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15306.)

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TABLE I, CLASSIFICATION AND BASIC REQUIREMENTS OF 4x2 LIGHT ELECTRIC TRUCKS

PARAMETER / TYPE/CLASS (PAYLOAD) (min.)	I/A (500 lb.)	II/A (500 lb.)	III/B (1000 lb.)	III/C (1500 lb.)	III/D (2000 lb.)	III/E (3000 lb.)
Application	Utility On Facility	Utility On Road	Delivery Van On Road	Delivery Van On Road	Delivery Van On Road	Delivery Van On Road
G.V.W.R. (maximum)	2000 lbs.	4500 lbs.	6000 lbs.	7500 lbs.	9000 lbs.	12,000 lbs.
WB / OL Ratio[1] (minimum-maximum)	.50 - .75	.50 - .75	.50 - .75	.50 - .75	.50 - .75	.50 - .75
Outside Width (maximum)	60"	66"	72"	72"	72"	80"
Rear Door(s) Opening w x h (inches) minimum	36" x 40"	36" x 40"	48" x 60"	48" x 60"	48" x 66"	48" x 66"
Cargo Space (minimum) ($ft.^3$)	25 $ft.^3$	50 $ft.^3$	150 $ft.^3$	225 $ft.^3$	350 $ft.^3$	500 $ft.^3$
Interior Height (minimum)	50"	50"	66"	66"	72"	78"
Approach/Departure Angles (minimum)	20°	20°	20°	20°	20°	20°
Front Door(s) minimum opening w x h	30" x 42"	30" x 42"	24" x 60"	24" x 66"	30" x 72"	30" x 72"
Occupants	2	2	2	2	2	2
Miscellaneous Data	see note (a)	see note (b)	see note (b)	see note (b)	see note (b)	see note (b)

[1] $\frac{\text{Wheel Base}}{\text{Overall Length}}$ $\frac{\text{W.B.}}{\text{O.L.}}$ Ratio

Note (a): Rear seating for one additional occupant optional (total 3 occupants)

Note (b): Rear seating for two additional occupants optional (total 4 occupants)

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3. REQUIREMENTS

3.1 Requirements. The vehicle(s) and the allied equipment furnished under this specification shall be provided from the manufacturer's current production. Vehicle(s) shall be completely equipped with all the necessary operating components and accessories customarily furnished the general public, or commercial market, whether stipulated herein or not, together with such modification and attachments as may be necessary or specified to enable the vehicle to function reliably and efficiently in sustained operation. The design of the vehicle and optional equipment shall permit accessibility for maintenance, replacement, and adjustment of component parts and accessories with minimal disturbance of other components and systems. The term "heavy-duty" as used to describe an item, shall mean in excess of the usual quantity, quality, or capacity that is normally supplied with the standard production vehicle or component. All vehicles covered by this specification must meet all applicable FMVSS requirements. Nothing written in this specification shall be construed to negate any FMVSS or any other safety requirement.

3.1.1.1 Battery module(s) accessibility. The vehicle shall be designed so that the propulsion battery module(s) can be serviced or removed without any body frame, or equipment obstructions. If the vehicle incorporates either a low or maintenance free propulsion battery, then the vehicle's body, frame, or equipment obstructions shall be minimized to provide access to the battery. Additionally, the vehicle's construction shall provide for the venting of battery fumes away from the vehicle's interior.

3.1.1.2 Battery retention and protection. The vehicle shall incorporate a battery restraint system which will prevent the battery from being dislocated in case of accident or rollover. Additionally, the battery shall be afforded protection against any road hazard or debris.

3.1.1.3 Motor and controller protection. The motor and controller shall be afforded protection against road hazards and debris.

3.1.2 Type I, Class A, Utility-on facility. The utility vehicle furnished shall be a closed style cargo/passenger compact utility vehicle (see 3.11.1). These vehicles shall be furnished with an integral windshield, doors with windows that can be opened in increments and locked when closed, individual or bench style seat(s) for driver and passenger. The enclosure shall be metal, plastic or fiberglass full cab-body provided with side panel and rear windows. When specified (see 6.2), rear seating for one additional occupant shall be provided. These vehicles shall meet the performance requirements in 3.7 for on facility vehicles.

3.1.3 Type II, Class A, Utility-on road. The vehicle shall be the same as Type I Utility (3.1.2), except the vehicle, when specified (see 6.2), shall contain seating for an additional two passengers in lieu of cargo space. Access to rear seating may be through a rear door (see 3.11.1). Additionally, these vehicles shall be capable of the performance requirements in 3.7 for on road vehicles.

3.1.4 Type III, Classes B, C, D, and E, Delivery van-on road. The vehicle furnished shall be a closed style van type vehicle with integral windshield (see 3.11.2). The vehicle shall have sliding front doors with windows that can be opened in increments and locked when closed. Unless otherwise specified, the vehicle furnished shall contain a front passenger seat, a rear cargo door and when specified (see 6.2), rear seating (see 3.10.4) for two occupants. The body shall be of metal, plastic, or fiberglass construction. Additionally, these vehicles shall be capable of the performance requirements in 3.7 for on road vehicles.

3.2 Operational requirements. The vehicle shall be capable of starting and operating efficiently under any of the conditions or combinations of conditions outlined herein, or as may be specified in the invitation for bid, contract, or order.

3.2.1 Temperature conditions. The vehicle shall be capable of meeting all performance requirements of paragraph 3.7 in ambient temperatures of 40°F to 90°F. The low temperature requirements shall be met after the specified low temperature has been sustained for not less than two days without benefit of solar radiation when serviced in accordance with the manufacturer's recommendations. The vehicle shall be capable of operating in any temperature between 0°F to +125°F (see 3.8.3.1.2). Additionally, the vehicle shall not be damaged by storage in temperatures from -20°F to +140°F for periods of at least four days duration, when the vehicle is serviced in accordance with the manufacturer's recommendations.

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3.2.2 Noise and sound level limits.

3.2.2.1 Exterior noise. The exterior noise level produced by the vehicle at the time of manufacture shall not exceed the Federal, State and local noise regulations applicable to where the vehicle will be based.

3.2.2.2 Interior noise. The interior noise level shall not exceed the maximum allowed under the requirements of the Federal Motor Carrier Safety Regulations (FMCSR).

3.3 Materials. Reclaimed materials shall be used to the maximum extent possible in both the vehicle and the batteries. All materials used on the vehicle shall be free from defects that affect the appearance, or impairs the form, fit, function, or reliability of any part, or the vehicle as a whole.

3.4 Vehicle, equipment and accessories. Except when otherwise specified, the vehicle, components, assemblies, bodies, devices, and accessories to be delivered under the contract shall be standard or optional commercial automotive, industrial, aircraft or military products, which meet or exceed the requirements of the specification. The vehicle shall comply with all Federal Motor Vehicle Safety Standards and Federal Regulations applicable to the specified vehicle on the date of manufacture. All chassis components and optional items shall be as represented in the manufacturer's current technical data, and mounted equipment shall be as represented in the equipment manufacturer's current technical data. Technical data shall be limited to specifications and technical materials, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in appropriate offices of the procuring activity prior to delivery of the item. The component parts of the vehicle need not be the products of the same manufacturer.

3.4.1 Heater/defroster and defogger. All electric vehicles furnished shall be equipped with a gasoline fueled heater/defroster and defogger that meet the requirements of SAE J 1024, in addition to all FMVSS and FMCSR performance and safety requirements.

3.4.1.1 Heater and defroster operation. The heater shall be capable of automatic ignition without the use of a pilot light and shall be capable of operating on any available grade of commercial gasoline. The system shall incorporate controls for temperature, selection of heater or defroster and three blower speeds.

3.4.1.2 Heater fuel storage. The fuel shall be stored in a tank which is physically located below the level of the vehicle's floor. The fuel shall be supplied by an electrically operated fuel pump. The fuel tank and delivery system shall meet all applicable FMVSS and FMCSR safety requirements. The access to the tank shall be from the exterior of the vehicle so as to preclude the entering of gasoline fumes into the vehicle. The fuel tank shall have ample capacity for 15 hours of continuous operation while operating under the severest conditions set forth in the Federal safety requirements for the defroster.

3.4.2 Ventilators. Vehicle shall be equipped with front mounted left and right fresh air vent, or equivalent. Vents shall be screened to prevent entry of foreign matter. When closed, the vents shall be weather tight. When specified (see 6.2), additional vents shall be installed at specified locations.

3.4.3 Battery chargers.

3.4.3.1 Battery charger-on-board. Unless otherwise specified (see 6.2), all vehicles furnished shall be equipped with an on board battery charger. The charger shall be capable of recharging the battery from an 80 percent discharged state to full charge in 10 hours maximum when operated on any voltage greater than 230V single phase. Additionally, when specified (see 6.2), the charger shall operate on 115 to 120V, 60Hz and recharge the battery from an 80 percent discharged state to full charge in 18 hours. The power cable that goes to the battery charger shall interface with a vehicle mounted connector that is accessible from the vehicle's exterior (in order to facilitate battery charging with the vehicle locked). Additionally, the power cable shall be capable of being secured to the vehicle's connector. The length of the power cable shall be as specified (see 6.2) in the invitation to bid, contract or purchase order.

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3.4.3.2 Battery charger-off-board. When specified (see 6.2), an off board (stationary) battery charger shall be provided. The charger shall be capable of recharging the battery from the 80 percent discharge state to full charge in 8 hours maximum, when operated on any voltage greater than 208V three phase or 230V single phase. Additionally, when specified (see 6.2), this charger shall be designed for outdoor use.

3.4.3.3 Charger interlock. When connected, all chargers shall automatically immobilize the vehicle's propulsion system to preclude the movement of the vehicle.

3.4.3.4 Charger shut off. All chargers shall include a provision that will sense the battery's state of charge and automatically shut off the charger when the battery is fully charged. The chargers output shall be cut off if the charger is overloaded or shorted. Additionally, when applicable, the charger shall also be interlocked so that it is inoperative unless it is connected to a battery having the correct voltage and polarity.

3.4.3.5 Ground fault interrupters. All battery chargers shall be equipped with a power line ground fault interrupter that meets all the National Electrical Code, state and local electrical code requirements.

3.4.3.6 Battery charger power requirements. All battery chargers furnished shall be capable of operating on the voltage, phase and frequency that is specified (see 6.2) in the invitation to bid, contract or purchase order. Unless otherwise noted, the specified power requirements shall not deviate from standard voltages and phases that are commonly available throughout the United States.

3.4.3.7 Charger status indicators. All chargers shall be equipped with an indicator which shows that the charger is energized, and an indicator which shows that the battery charger has shut down because the battery has been fully charged.

3.4.3.8 Charger temperature requirements. All battery chargers shall be capable of operating as specified under any ambient temperature from 0°F to 125°F.

3.5 Extra or optional equipment. In addition to the standard vehicle and components specified, each vehicle shall be furnished with optional or special equipment as specified (see 6.2). All equipment furnished and provided, unless otherwise specified, shall be mounted and installed ready to operate.

3.6 Vehicle weight, payload, ratings, and dimensions.

3.6.1 Curb weight. The curb weight of a vehicle is defined as the weight in pounds, of the complete vehicle. Curb weight shall include the chassis, cab and body, all permanently attached devices, equipment, full complement of lubricants, batteries and charger (mounted on vehicle).

3.6.2 Payload capacity, minimum. Minimum payload capacity specified in 1.2.1 is defined as the vehicle's useful carrying capacity in pounds (lbs.;#). The payload shall include the weight of the cargo, and the occupants (at 150 pounds per occupant). The specified payload may be decreased in the amount equal to the weight of the extra equipment specified and installed such as racks, utility cabinets, etc. In no event shall the payload requirement be decreased by more than 30 percent. The payload in pounds shall be calculated as the difference between the G.V.W.R. of the class vehicle furnished less the curb weight (see 3.6.1). The difference shall be termed the net payload.

3.6.3 Gross vehicle weight rating (G.V.W.R.). The vehicle gross weight rating shall be equal to or greater than the combination of weights of the vehicle's curb weight (3.6.1) and the minimum payload specified (see 3.6.2). Manufacturers shall provide a rating label showing the actual G.V.W.R. of the vehicle furnished.

3.6.4 Dimensions. Vehicle dimensional requirements are specified in Table I for each type and class vehicle. The OL (overall vehicle length) and wheelbase dimensions shall permit the distribution of weight to allow for proper steering control, and traction for the drive wheels under all operating conditions.

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3.7 Performance.

3.7.1 Performance requirements. Unless otherwise specified (see 6.2), the following minimum performance requirements in Table II (page 9) are applicable to all vehicles as specified.

3.7.2 Performance test requirements. The test procedures of SAE J227a shall be used to determine compliance with the performance requirements of Table II.

3.8 Chassis, chassis components.

3.8.1 Chassis, frame. The chassis-frame shall be capable of withstanding the strains of on-off road service or any special service and equipment requirements as specified.

3.8.2 Vehicle lubrication. The chassis components, devices, accessories, and equipment requiring lubrication maintenance shall have provision for lubrication.

3.8.3 Power unit - motor, controller, and battery.

3.8.3.1 General requirements.

3.8.3.1.1 Electrical requirements. The entire propulsion electrical system (motor, controller, battery and wiring) shall be completely isolated from the vehicle's chassis ground. Additionally, the propulsion system shall be protected against shorts and overloads by means of a circuit breaker or fuse. The circuit breaker or fuse shall be located so that the vehicle's operator can easily reset the circuit breaker or replace the fuse from the driver's seat.

3.8.3.1.2 Temperature operating range. The propulsion motor, controller and battery shall be capable of continuous operation at any ambient temperature from 0°F to 125°F without any degradation in reliability. All motors, controllers and batteries shall be capable of operating continuously without overheating, tripping circuit breakers, or blowing fuses when meeting any of the performance requirements of 3.7 or any other normal operational condition with the vehicle at its G.V.W.R.

3.8.3.1.3 Overheating indicator. Both visual and aural indicators shall be employed to indicate when the motor and/or controller are operating above the normal maximum temperature.

3.8.3.2 Motor requirements.

3.8.3.2.1 Motor construction. The propulsion motor shall be constructed using class F or better insulation.

3.8.3.2.2 Motor lubrication. If the bearings require periodic lubrication, a means to lubricate them without motor disassembly shall be provided.

3.8.3.2.3 Motor brush replacement. The motor shall be constructed so that the periodic brush inspection, and when necessary, brush replacement can be accomplished without removing or relocating any other part of the vehicle or accessory except a dust or water-proofing cover.

3.8.3.3 Controller requirements.

3.8.3.3.1 Controller function. The controller shall provide smooth, efficient and safe control of the motor and vehicle. It shall provide power to the motor in proportion to the deflection of the accelerator pedal. The controller shall also limit the power to the motor in order to prevent it from producing excessive torque which could damage the motor, transmission (if so equipped) and drive train. The forward, reverse and neutral positions may be obtained by use of a switch, or by the transmission. The controller shall incorporate an emergency power disconnect switch, which will remove all power from the motor and controller. The switch must be readily accessible from the driver's seat and must be easily recognized. Additionally, the controller shall also incorporate circuits which automatically provide for regenerative braking (see 3.8.7.1).

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PERFORMANCE REQUIREMENTS

TABLE II

PERFORMANCE REQUIREMENTS (BASED ON THE SAE J227a TEST PROCEDURE)	TYPE I (ON FACILITY)	TYPE II & III (ON ROAD)
Range (Miles) at Constant Speed per SAE J227a		
20 MPH	60 Miles	-
25 MPH	55 Miles	75 Miles
30 MPH	45 Miles	70 Miles
35 MPH	-	60 Miles
40 MPH	-	50 Miles
45 MPH	-	35 Miles
Selected Driving Patterns Cycles		
SAE J227a - Schedule A	1500	-
SAE J227a - Schedule B	225	250
SAE J227a - Schedule C	75	100
SAE J227a - Schedule D	-	45
Acceleration[1] (Seconds) per SAE J227a		
0-30 MPH	15 seconds	10 seconds
0-40 MPH	-	17 seconds
Gradeability Limit[1] (Percent Grade) per SAE J227a	25%	30%
Gradeability at Speed[1] MPH/Percent Grade per SAE J227a	25 MPH[2] / 3%	40 MPH[3] / 3%

1 with the battery discharged 80% (based on specific gravity readings)

2 for a distance of not less than 1 mile

3 for a distance of not less than 2 miles

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3.8.3.3.2 Controller interlock requirements. Application of either the parking brake or service brakes shall automatically inhibit power from being applied to the motor. Additionally, the transition from braking to power must be accomplished smoothly and without lapse to prevent rolling during stop-start operation on hills.

3.8.3.3.3 Controller - design requirements. The controller shall be designed to provide long service life. Unless otherwise specified, the controller shall incorporate hermetically sealed (military type) semiconductors. Any relays or contactors that are not hermetically sealed shall have the coils encapsulated with an epoxy or equivalent type encapsulant. All electrical and electronic components that are part of the controller shall be selected so that they have reserve ratings of at least 25 percent, in terms of voltage, current and power. These components shall also have temperature ratings of at least 25 percent higher than the worst case component temperature encountered under the highest ambient operating temperature as per paragraph 3.8.3.1.2.

3.8.3.3.4 Electromagnetic interference. The vehicle shall not exceed the maximum radiated noise requirements of SAE J551.

3.8.3.4 Battery, propulsion.

3.8.3.4.1 Battery construction. The battery shall be constructed so that over the life of the battery, vibration or any other factor encountered either in operation, charging or placing the vehicle in any position encountered in normal operation, shall not cause the case or cell caps to leak acid or any other electrolyte.

3.8.3.4.2 Battery maintenance. The battery shall either be maintenance free or incorporate a system for adding fluid to the electrolyte from a single location.

3.8.4 Transmission. If the vehicle is equipped with a transmission, and gear changes are necessary to meet all the performance requirements of 3.7 and 3.8.7.2, the gears shall be capable of being changed while the vehicle is in motion.

3.8.5 Axles.

3.8.5.1 Axle, ratios. Unless otherwise specified (see 6.2), the driving axle shall be equipped with the numerical ratio differential which offers the greatest energy economy consistent with the performance requirements as specified in 3.7.

3.8.6 Suspension. Vehicle shall be equipped with the heaviest-duty suspension offered for the model specified. When specified (see 6.2), auxiliary, 2-stage, or helper rear springs shall be furnished having a rated capacity of at least 10 percent over the primary spring requirements. Additionally, the vehicle shall be capable of operating under any normal road conditions without any undesirable sway, leaning or instability.

3.8.6.2 Spring stops. The manufacturer's standard spring axle stops shall be furnished. The stops shall prevent the axles from striking any part of the motor, controller, accessories or undercarriage.

3.8.6.3 Shock absorbers. Shock absorbers, double-acting type, heaviest-duty available for model offered shall be furnished on the front and rear axles of all vehicles.

3.8.6.4 Steering. Unless otherwise specified (see 6.2), manual mechanical type steering shall be furnished.

3.8.7 Brakes.

3.8.7.1 Braking systems. In addition to the manufacturer's standard brakes, all vehicles shall be equipped with a regenerative braking system.

3.8.7.2 Regenerative braking system. The regenerative braking system shall be activated in proportion to the force applied to, or the displacement of the service brake pedal. The system shall be capable of limiting the speed of the vehicle when loaded to its G.V.W.R. to 30 MPH on a 10 percent downgrade, for a distance of not less than one mile. Additionally, when the system is activated, it shall provide regenerative braking in a smooth, jerk-free manner.

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3.8.8 Wheels, rims, tires and inner tubes.

3.8.8.1 Wheels, rims. The manufacturer's heaviest-duty single wheels and rims shall be provided. The wheels and rims shall conform to the recommendations of the Tire and Rim Association, Inc. Dual rear wheels shall be furnished on vehicles over 9,000 lbs G.V.W.R.

3.8.8.2 Tires.

3.8.8.2.1 Tires. Unless otherwise specified (see 6.2), tires shall be tubeless with regular highway tread. The tire size, type, tread and load range shall be the same on all wheels of the vehicle, including the spare. The load limit of the tires shall not be increased by specifying a maximum speed limit of the vehicle.

3.8.8.2.2 Mud and snow tires. When specified (see 6.2), mud and snow tread for the tires shall be provided on the drive wheels.

3.8.8.3 Inner tubes. When tube type tires are furnished, inner tubes shall conform to the recommendations of the Tire and Rim Association, Inc.

3.8.8.4 Spare wheel and tire. One spare inflated rim/tire assembly shall be furnished conveniently accessible in a suitable carrier. Carrier design shall enable removal and installation of spare wheel assembly using only the tools specified in 3.9.8.2. If the spare is externally mounted, a suitable locking device shall be provided.

3.8.8.5 Tire chain clearance. Tire chain clearance shall be provided in accordance with SAE J683a.

3.8.8.6 Wheel-tire balancing. All wheel-tire assemblies of the vehicle, including the spare, shall be balanced in accordance with current manufacturing standards.

3.8.9 Electrical system, accessory.

3.8.9.1 Electrical system and equipment. A 12 volt electrical system which includes a converter and 12 volt battery, shall be provided for all lighting and 12 volt accessories. The accessory system shall be protected with the necessary circuit breakers or fuses which are readily accessible.

3.8.9.2 12 volt converter system. The converter system shall be capable of constantly supplying the normal steady state 12 volt power requirements of the entire vehicle. The peak or surge requirements (e.g., stop lights, directional signals, etc.) may be supplied by the accessory battery. The time required to recharge the 12 volt accessory battery shall not be greater than the recharge period for the main propulsion battery.

3.8.9.3 Battery, accessory.

3.8.9.3.1 Battery, accessory capacity. The accessory battery provided shall have a minimum performance rating of 70 A.H. at the 20 hour rate.

3.8.9.3.2 Battery type. This battery shall be of low-maintenance or maintenance-free construction.

3.9 Cab-body, equipment and accessories.

3.9.1 Cab-body. A conventional integral cab-body shall be provided with the following minimum equipment: Manufacturer's standard insulation, sound deadening materials, floor covering, front doors with opening and locking windows, rear doors, and when specified (see 6.2), a side cargo door. All doors shall have checks and stops, locks operable from the inside, front doors equipped with external key-locks, two sets of keys, glove (dispatch) compartment, heaviest duty vinyl upholstery available of the manufacturer's standard color shades or mixtures, except solid black or white, compatible with exterior color, full-width adjustable seat with back, or when applicable, individual driver and helper seats (minimum seating of two per vehicle), and rear body window(s). The body shall be adequately sealed to prevent entry of water and dust.

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3.9.2 Bumpers. A sturdy front, full-width bumper shall be provided for each vehicle type. "Full-width" is defined as being at least 95 percent of the frontal area width. Rear end protection or bumpers conforming to Federal Motor Carrier Safety Regulations, as applicable to the vehicle and body offered, shall be furnished on all vehicles. Bumpers shall not be fastened to rear fenders nor to the sides of the body.

3.9.3 Controls and operating mechanism. Unless otherwise specified (see 6.2), controls and operating mechanism shall be located for left-hand drive. Equipment controls shall be complete and conveniently located for the driver. Lever controls for specified equipment shall be designed and located to permit easy entrance and exit of the operator to and from driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing.

3.9.4 Seats. All seats shall be the manufacturer's heaviest duty seat assemblies for the model offered. The upholstery shall be as specified in 3.9.1.

3.9.5 Seat and shoulder belts. The driver and front passenger shall be provided with seat and shoulder belts, with inertia locking or enclosed retractor devices. All other passenger seats shall be equipped with lap type seat belts of the inertia locking type.

3.9.6 Glass. Unless otherwise specified (see 6.2), glazing shall be clear.

3.9.7 Instrument panel. The instrument panel shall include, but not be limited to, indicator lights, indicating speedometer and recording odometer which meets the SAE J678 accuracy requirements, lighting switches, heater-defroster controls, motor temperature indicator and battery state of charge meter. Additionally all controls and instruments shall be illuminated.

3.9.7.1 State of charge meter. The state of charge meter shall be capable of indicating the useful remaining battery capacity from the fully charged state, in fractional terms, e.g., full, 3/4, 1/2, 1/4, 0. The meter should read "0" when the battery is discharged to the lowest level (e.g., 80 percent) recommended by the manufacturer (typically 80 percent discharge). The accuracy of the "state of charge" system shall be within $\pm$ 5 percent, but in no case shall the meter indicate above "0" when the battery is at it's lowest "state of charge" recommended by the manufacturer. The meter pointer shall remain steady and unaffected by meter current transients.

3.9.8 Accessories, cab body. Unless otherwise specified, the accessories and convenience items and devices furnished shall include, but not be limited to, fresh-air type heater with defroster (see 3.4.1), dual sun visors, dual windshield wipers and washers, inside rearview mirror, outside rearview mirror (see 3.9.8.1), outside clearance reflectors when required, tools (see 3.9.8.2), and mud flaps (see 3.9.8.3), when applicable.

3.9.8.1 Mirrors, outside. Dual western or low-level style outside rear view mirrors as applicable to the body specified shall be furnished. Mirrors shall have a minimum of 12 square inches reflective area each for Type I vehicles, and 60 square inches of reflective area each for other type vehicles. Large western style mirrors and mounting brackets may be shipped dismounted within the vehicle.

3.9.8.2 Tools and tool storage. Each vehicle shall have a jack of sufficient capacity to easily lift the full-loaded vehicle to the height necessary to permit removal and replacement of wheel and tire assembly. A jack handle, and wheel lug-nut wrench, shall also be included. Sufficient storage shall be provided for furnished tools.

3.9.8.3 Mud flaps. Mud flaps shall be furnished on all vehicle bodies, when the rear wheels and tires are exposed above the centerline of the rear axle. Mud flaps furnished shall be full-width of rear tire(s), of appropriate ground clearance, twist and sail proof, and fastened securely, conforming to SAE J682, and devoid of advertisement except for supplier's name or emblem.

3.10 Truck body design. Unless otherwise specified by authoritative documents, each truck body shall be designed and mounted to fit the vehicle chassis, which shall provide the proper weight distribution for steering and balance, and present a satisfactory appearance. Body widths shall be not less than the overall width of the rear

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axle plus tires. Bodies designed with wheel openings shall have the vehicle's wheels centered within 2 inches. Materials used in the fabrication of bodies, if not specifically described, shall be not less in quality and strength than those materials normally used by the most reputable manufacturers.

3.10.1 Types I and II, Class A. When specified (see 6.2), the utility vehicle shall be a cargo/passenger compact body and cab. These vehicles shall be furnished with an integral windshield and doors with cranked or sliding glass windows than open, heavy duty individual or bench style seat(s) for driver and passenger, and cab-body enclosure. The enclosure shall be metal, fiberglass or plastic full cab-body, of one piece construction provided with side and end panel window. Rain gutters shall be integrated with roof panels and be located over all doors and opening windows. Each enclosure side window shall be capable of being opened with an opening of not less than 95 square inches. The windowed rear door or tailgate shall be retained in the open position with stays. When specified (see 6.2), heavy duty rear seat(s) shall be furnished, with front seat(s) designed to permit easy ingress and egress. Utility vehicle dimensions shall be as shown in Table I.

3.10.1.1 Interior. Body sides shall be lined and reinforced for load support. The roof shall be insulated and lined with one inch fiberglass batts, or the equivalent. The side and roof lining shall be capable of withstanding the severe usage normally encountered in utility vehicles. At least one dome light shall be provided which is capable of illuminating both the driver and rear cargo areas.

3.10.1.2 Floors. All floors in the cargo shall be constructed to withstand the severe usage normally encountered in utility vehicles. The floors in the occupants' area shall be entirely covered with the manufactuer's heavy duty rubber covering.

3.10.2 Type III, Class B, C, D, and E, delivery van body. When specified (see 6.2), van walk-in bodies shall be built on a forward control chassis of the class specified in Table I. It shall have an integral driver's compartment of a unitized construction. Heavy duty upholstered seats for the driver and passenger shall be provided. Driver's seat shall be adjustable, fore and aft, up and down, with a positive locking device. The seat shall be capable of being adjusted without the use of any tools or special equipment. When specified (see 6.2), upholstered rear seating for two occupants shall be provided. Van dimensions shall be shown in Table I.

3.10.2.1 Body, Type III. Except as otherwise specified herein, the body shall be constructed of sheet metal having minimum gages specified for carbon steel; high tensile steel utilized may be two gages lighter weight in accordance with U.S. Standard gages. Aluminum, magnesium alloys, fiberglass or plastic of equal or greater strength may be furnished. Inset steps shall be provided at both front entrances. Body shall have sliding type front doors with sliding windows. Double rear doors, or when specified (see 6.2), a single or double rear roll-up door shall provide a rear centered opening. Rear doors shall have door operating handles inside and outside. Rear doors furnished shall be provided with door checks to hold door(s) in full open position. In addition to the door operating handles, grab handles shall be provided on the inside of the rear door(s). All doors shall be furnished with safety glass in upper section, and with standard locking arrangement. A safety tread, steel plate step, integrated with the rear bumper and extending not less than the width of door opening, shall be furnished. Rear end splash shields shall be furnished extending from ends of steps to outside edges of body.

3.10.2.2 Framing. Not less than five full width crossmembers and wheelhouse short crossmembers on maximum 24-inch centers (including end sills) shall be furnished. Stud crossmembers, as required at wheelhousing to provide required strength shall be furnished.

3.10.2.3 Side panels and roof. Side panels shall be constructed of stretcher level or formed sheet steel of not less than 22 gage above and 16 gage below the belt line of the body. Aluminum, magnesium alloys, fiberglass or plastic of equal or greater strength and durability may be substituted for the side panels or roof. Rain gutters shall be integrated with roof panels and be located over all doors and opening windows. All body seams and joints shall be sealed against weather and dust.

3.10.2.4 Floor. Floor shall be wood and metal, all wood, or all metal construction. For wood and metal floor, wood flooring shall extend the f-l length of loadspace and the width of loadspace between the wheelhousing; not less than 16 gage (0.055 inch) steel shall be utilized for remainder of floor area. Wood flooring shall be not less than 3/4-inch thick planking or 1/2-inch laminated exterior grade plywood, metal covered with not less than 24 gage (0.0239 inch) steel, provided with recessed steel

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skid strips, installed flush with ends of flooring and protruding not more than 1/16-inch above floor surface. All metal floors, where smooth sheet steel is used, shall be not less than 11 gage (0.1196 inch) except for vehicles with unitized body construction which shall be not less than 14 gage (0.0747 inch); or when floor is of ribbed sheet steel, floor shall be not less than 18 gage thick. All metal floor, when safety tread sheet steel is used shall be not less than 12 gage (0.105 inch), except for the driver's area and steps which shall be not less than 16 gage (0.055 inch). Floor and support members shall sustain an evenly distributed load of 150 pounds per square foot of floor area.

3.10.2.5 Interior. Body sides shall be lined and reinforced for load support. Roof shall be lined and insulated with not less than 1-inch fiberglass batts. Headliner for roof, and side lining shall be sheet metal, 1/8-inch tempered masonite, or 1/4-inch plywood. When plywood is furnished, exposed side shall be free from defects and smooth sanded. All interior side panels shall be removable to facilitate body repair. At least two dome lights shall be installed, one in cargo and one in the driver's compartment. Minimum dimensions shall be as specified in Table III.

3.11 Additional or special requirements.

3.11.1 Special requirements. When specified (see 6.2), this section specifies items which may be required because of unusual operating conditions. If these items are in conflict with the manufacturer's standards and options, those specified herein shall take precedence.

3.11.2 Overseas vehicle requirements. When specified (see 6.2), the following equipment changes, items, devices, and adjustments to systems shall be furnished with right hand drive.

(a) Left dip headlights (as used on right-hand drive vehicles) adjusted for driving on the left side of the road (prior to shipment).

(b) Speedometer-odometer, calibrated in kilometers.

(c) Defogging fan. If heater-defroster is deleted or not furnished, fan shall be mounted on the dashboard convenient to the driver.

(d) Controls, right-hand drive (vehicles).

(e) Tube type tires.

(f) Battery charger to operate on voltage and frequency as specified.

3.11.3 Rustproofing. When specified (see 6.2), vehicle(s) shall be rustproofing in accordance with Fed. Std. No. 297 for civil agencies, and Mil. Std. 1223 for military vehicles.

3.11.4 Undercoating. When specified (see 6.2), undercoating shall be applied in accordance with Mil. Std. 1223.

3.11.5 Markings, data plates, warranty notice, etc. Unless otherwise specified (see 6.2), as cited by the procuring activity for the appropriate military service, identification markings, data plates, and warranty notice shall be provided, conforming with Mil. Std. 1223. Civil agencies' vehicles shall be provided with concealed data plates and warranty notice in accordance with Mil. Std. 1223.

3.12 Preparation for painting and color.

3.12.1 Preparation for painting. The bodies and furnished equipment surfaces concealed or exposed, except polished metal parts shall be cleaned, treated, and coated with a firm primer and preservative with rust inhibiting properties in accordance with Mil. Std. 1223. Subsequent finishes shall be industry's standard practice.

3.12.2 Paint, finish or color. Unless otherwise specified (see 6.2), the civil agencies vehicles exterior color shall be the manufacturer's standard nonmetallic light color, except red and black. When specified (see 6.2), finish color shall match color chip or shade of Fed. Std. No. 595. The interior finish shall be manufacturer's standard compatible color with the exterior. Unless otherwise specified (see 6.2), body and exterior mounted equipment shall be the same color as the cab.

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Military vehicles treatment, painting and color(s) shall be in accordance with Mil. Std. 1223, as specified by the procuring activity for the appropriate service.

3.13 Pre-delivery inspection and servicing. Prior to authorized acceptance of the vehicle(s), the contractor shall service and inspect each vehicle in accordance with manufacturer's prescribed pre-delivery servicing requirement or GSA Form 1455, as applicable, and apply Government supplied signed verification label. Servicing shall also comply to ambient temperatures and conditions applicable with the route to transport to the consignee and ultimate destination (see 5.1).

3.14 Workmanship. Vehicles shall be free from defects which may impair their serviceability or detract from appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. The contractor is obligated to prove that the vehicle, components, and equipment offered/provided meets or exceeds the requirements of the contract. The contractor may use his own, or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to the prescribed requirements. The contractor shall provide the Government inspection representatives, instruments, equipment, and such assistance as they may find necessary for the conduct of checks, tests or inspections.

4.2 Classification of inspections. The contract or purchase order shall specify (see 6.2), the inspection(s) required for each vehicle. The inspections are classified as follows:

(a) First production vehicle inspection and test (see 4.3 thru 4.3.2, 4.4, thru 4.4.4, and 4.5)
(b) Quality conformance inspection (see 4.7 thru 4.8)

4.3 First production vehicle inspection. Unless otherwise specified (see 6.2), the first production vehicle of each type specified under contract shall be inspected in accordance with the examination in 4.3.2 and all tests under 4.4. The first vehicle produced under the contract shall be completely representative of the balance of the vehicles to be delivered under the contract. The purpose of the inspection shall be to determine vehicle conformity with the contract. The vehicle shall be inspected by the contractor at his plant under the direction and in the presence of a Government representative(s). Inspections shall be conducted during normal daytime working hours. The cost of inspections, examination, all tests, and vehicle support, exclusive of personal expenses incurred by participating Government representatives shall be borne by the contractor. Acceptance of the first production vehicle shall not constitute a waiver by the purchaser of their rights under the provisions of the contract.

4.3.1 Vehicle weights. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axle. The imposed loading on front and rear axle will be computed using the curb weight and the payload as specified in 3.6. Calculated imposed loads on front and rear axle will be utilized to ascertain that suspension, axles and tires furnished are of adequate capacity to meet contract requirements.

4.3.2 Examination. The first production vehicle shall be examined to determine compliance with the requirements in Section 3 (except 3.7).

4.4 Tests.

4.4.1 Test criteria and certifications. The vehicle shall be prepared for operation in accordance with the manufacturer's recommendations. The vehicle shall be manned and loaded to simulate maximum payload conforming to its class. The road test course shall be approved by the procuring activity, or Government representative.

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The substitution of a new vehicle or replacement of major component(s) may require complete retest at the discretion of the Government representative. To avoid duplication, certification(s) may be acceptable in lieu of actual testing, providing authoritative verifying data is on file for examination. The certifications will remain valid as long as they are applicable to the vehicle model, component and equipment specified, so offered and furnished. Certifications that appear on the vehicle need not be resubmitted (i.e., Department of Transportation, DOT). Certifications with verifying test data for Federal procurements shall be forwarded to the Office of Standards and Quality Control (FMQ), General Services Administration, Federal Supply Service, Washington, DC 20406, and shall also be provided on request to other authorized agents.

4.4.2 Performance tests. The vehicle and equipment as furnished shall be examined or tested to verify conformance of the following:

(a) Vehicle physical dimensions (see 1.2, Table I)
(b) Battery accessibility (see 3.1.1.1)
(c) Battery retention and protection (see 3.1.1.2)
(d) Motor and controller protection (see 3.1.1.3)
(e) Operating temperature conditions (see 3.2.1)
(f) Heater/defroster and defogger performance (see 3.4.1)
(g) Battery charger performance and requirements (see 3.4.3)
(h) Performance requirements (see 3.7 and Table II)
(i) Provisions for lubrication (see 3.8.2 and 3.8.3.2.2)
(j) Performance of regenerative braking system (see 3.8.7.2)
(k) Spare wheel and tire, and if necessary, locking device (see 3.8.8.4)
(l) Performance of 12 volt converter system (see 3.8.9)
(m) Water spray test (see 3.9.1 and 4.4.3)
(n) Performance of state of charge meter (see 3.9.7.1)
(o) Tire changing tools and jack performance (see 3.9.8.2)
(p) Road test (see 4.4.4)

4.4.3 Water spray test. The vehicle shall be subjected to a water spray test for a minimum of 15 minutes. The spray shall be delivered by nozzles operating at a minimum of 25 pounds per square inch (psi) water pressure, sufficient in number and placed (approximately 3 feet from the body) to afford full coverage of sides, roof, front, rear and under carriage of the vehicle. Vehicles undergoing quality conformance inspection (4.7) shall be subjected to a water spray test for not less than 2 minutes duration. Evidence of water leakage shall be cause for rejection until leaks are corrected.

4.4.4 Road test. The vehicle shall be subjected to a minimum 40 mile road test of which 20 miles shall be continuous miles on paved roads at a minimum speed of 30 MPH for Type I vehicles, a minimum speed of 45 MPH for Type II & III vehicles, and a distance of 20 miles under stop and go urban conditions for all three vehicle types. The vehicle shall be tested to determine if it performs smoothly, handles in a stable manner, is free of noise, vibration, and any other undesirable traits. Additionally, all battery cells shall be tested for conformance to the battery manufacturers' specifications. After completion of the road test, vehicle shall be subject to the water spray test.

4.5 Inspection failure of first vehicle. Failure of first production vehicle to meet all requirements and successfully complete examinations and test in 4.4 shall be cause for nonacceptance of any of the contract quantity, until the deficiencies and evidence of the corrective action is taken which precludes the recurrence of similar deficiencies. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries. Additionally, nonacceptance shall be for deficiencies including, but not limited to the following:

(a) Damage caused by collision.
(b) Failure of any major component.
(c) Vibration due to misalignment of wheels, frame, driveshaft, etc.
(d) Vibration due to type of body construction or mounting.
(e) Evidence of abnormal tire wear due to misalignment or unbalanced wheels/tires.
(f) Failure of any vehicular safety device such as brakes, steering assembly, windshield washers and wipers, or electrical circuits.
(g) Evidence of structural weakness in any part of the vehicle, vehicle components, or accessories.
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

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4.6 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract. Vehicle shall be then cleaned, serviced, and refurbished to the extent required for delivery of a new vehicle to the purchaser.

4.7 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to all production vehicles offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (d) as follows:

(a) Workmanship inspection (see 3.14)
(b) Vehicle operational checks (see 4.7.1)
(c) Water spray test (see 4.4.3)
(d) Battery cell check, to manufacturers' specifications

4.7.1 Operational checks. Operational checks of the vehicle shall cover all controls, electrical systems and devices, doors, windows, equipment, accessories, in and outside the vehicle. The vehicle shall be driven at all speeds, turns made at minimum radii, brakes tested for dependability, checked for rattles and squeaks, and compliance to 3.13. All controls, and mechanisms shall function and operate as intended.

4.8 Failure of inspection. The failure of any unit to pass an examination or test shall be cause to refuse acceptance of the vehicle until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency.

5. PREPARATION FOR DELIVERY

5.1 Preparation. Unless otherwise specified (see 6.2), the vehicle(s), shall be preserved and packaged for mobile delivery in accordance with the supplier's standard commercial practice, ensuring carrier acceptance and safe delivery to the destination in compliance with regulations applicable to the mode of transportation. In addition, the batteries shall be charged sufficiently to drive the vehicle a minimum of 30 miles under mixed driving conditions.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general administrative use by the Government in transporting personnel or cargo. Principal use of these vehicles will be for maintenance, parcel and/or mail delivery, messenger service, and routine security patrol. These vehicles are designed for on facility use, and stop and go low speed urban/suburban applications. (See the boxed note on Page 1 and Paragraph 1.1.1).

6.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government. Detailed descriptions, refinements, and deviations of each vehicle type are defined within this specification but may be modified in the invitation for bids or contractural requirement.

6.1.2 Precautions and observations. Due to the variety of truck models covered in the specification, purchasers should be aware that many equipment and accessory items described herein are not necessarily compatible or available on all vehicles. It should be ascertained before specifying (ordering) that vehicle manufacturers offer such equipment, devices, or options. (Reference manufacturer's data books and brochures). The parameters for Government agencies to purchase these vehicles are covered in the Federal Property Management Regulations (FPMR).

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6.2 Ordering data. Purchasers should select the equipment and options permitted herein and include the following information in the procurement document.

(a) Title, number and date of this specification and amendment number, if any.
(b) Quantity, type and class of vehicle required (see Table I and Paragraphs 1.2, 3.10.1, and 3.10.2).
(c) Rear seat(s); specify if required (see 3.1.2, 3.1.3, 3.1.4, 3.10.1, 3.10.2).
(d) Ventilators; specify additional vents and their locations (see 3.4.2).
(e) Battery chargers; specify on or off board chargers (indoor or outdoor) and line voltage, single or 3-phase, current limit, frequency, type of plug configuration and cable length (see 3.4.3 - 3.4.3.7).
(f) Performance; specify if requirements differ from Paragraph 3.7.
(g) Axle ratios; specify any unusual operating conditions which may require different axle ratios (see 3.8.5.1).
(h) Helper springs; specify when required (see 3.8.6.1).
(i) Power steering; specify when required (see 3.8.6.4).
(j) Tube type or other than highway tread tires; specify when required (see 3.8.8.2.1).
(k) Mud and snow tires; specify when required (see 3.8.8.2.2).
(l) Tinted windows; specify when required (see 3.9.6).
(m) Side cargo door; specify when required (see 3.9.1).
(n) Right hand drive; specify when required (see 3.9.3).
(o) Glass; specify tinted glass when required (see 3.9.6).
(p) Rear doors; single or double rear roll-up door, specify when required (see 3.10.2.1).
(q) Overseas requirements; specify when required (see 3.11.2).
(r) Rustproofing; specify when required (see 3.11.3).
(s) Undercoating; specify when required (see 3.11.4).
(t) Markings, data plates, warranty notice, etc.; specify if other than standard (see 3.11.5).
(u) Paint; specify color (see 3.12.2).
(v) Equipment color; specify if other than body color (see 3.12.2).
(w) Inspections; specify required inspection (see 4.2).
(x) Preparation for delivery; specify if other than standard preparation is required (see 5.1). Also state mode of delivery preferred.
(y) Shop repair manual(s) and parts list or book; specify when required (see 6.5).

6.3 Procurement requirements. Invitation for bids, contract, or orders should contain the following contractual requirements in Paragraphs 6.4 through 6.7, except in those instances where it is determined that inclusion thereof would not be to the best interest of the Government.

6.4 Warranty.

6.4.1 Warranty coverage. The contractor shall warrant the vehicle and furnished equipment against parts failure and malfunction due to design, construction or installation errors, defective materials or workmanship, and missing or incorrect parts (exceptions, see 6.4.4). The terms of the warranty shall be as per 6.4.1.1, 6.4.1.2, 6.4.1.3 and 6.4.1.4. This shall be exclusive of any authorized accumulated driveaway mileage. However, if the contractor receives from any supplier or subcontractor additional warranty on the whole or any component of the vehicle, in the form of time and/or mileage, including any pro rata arrangements, or the contractor generally extends to his commercial customers a greater or extended warranty coverage, the Government shall receive corresponding warranty benefits. The warranty commences when the Government accepts the vehicle from the contractors FOB point of origin/destination (see 3.13).

6.4.1.1 Electrical system. The motor, controller, state of charge indicator, battery charger(s), wiring, connectors and any other part of either the propulsion or 12 volt accessory system shall be warranted for 3 years or 36,000 miles, whichever comes first.

6.4.1.2 Propulsion battery. The propulsion battery shall be warranted for 36 months, 36,000 miles or 750 charge/discharge cycles, whichever comes first. The warranty for the last 18 months, 18,000 miles or 375 charge/discharge cycles shall be on a pro rata basis.

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6.4.1.3 Accessory Battery. The accessory battery shall be warranted for a minimum period of 36 months.

6.4.1.4 Balance of Vehicle. The balance of the vehicle shall be warranted for minimum period of two years, or 24,000 miles, whichever comes first.

6.4.2 Domestic use. When vehicles are used within the fifty States of the United States, the District of Columbia, Puerto Rico and the Virgin Islands, the warranty shall include the furnishing, without cost to the Government (FOB contractor's nearest dealer or branch to vehicle's location or station) new parts and assemblies to replace any that failed or malfunctioned within the warranty period. In addition, when the Government elects to have the work performed at the contractor's plant, branch, dealer, or contractor approved facility, the cost of the labor involved in the replacement of the failed or malfunctioned parts or assemblies shall be borne by the contractor.

6.4.3 Foreign use. When vehicles are used outside the fifty States of the United States, the District of Columbia, Puerto Rico, and the Virgin Islands, the warranty shall include the furnishing of new parts or assemblies to replace any returned to the contractor by the Government which failed or malfunctioned within the warranty period. The replacement parts or assemblies shall be delivered by the contractor to the port of embarkation in the United States designated by the Government. The contractor shall not be required to bear the cost of the labor involved in correcting defects in vehicles operated in foreign countries.

6.4.4 Warranty exceptions. Unless within the additional coverage under 6.4.1, the following items are considered normal maintenance and repair for which the contractor need not assume liability for reimbursing the Government regardless of the vehicle age or mileage.

(a) Abuse, negligence, or unapproved alteration of original parts.
(b) Damage from accidents.
(c) Brake linings or adjustments.
(d) General tightening, head lamp adjustments.
(e) Wheel alignment or tire balancing.
(f) Tires (if warranted by their manufacturer).
(g) Lamp replacement (unless caused by a malfunctioning electrical system).
(h) Miscellaneous expense such as towing, telephone, travel, lodging, or loss of personal property.

6.5 Operators, servicing and parts manuals. The successful bidder shall furnish at least one operator's and maintenance handbook, including a handbook(s) for any furnished special equipment. When specified (see 6.2), parts list or book and shop repair manual(s) for the vehicle and equipment furnished shall be provided.

6.6 Repair parts and service. As continuous operation of the vehicles contemplated by this specification is of utmost importance, it is necessary that the successful bidder be in a position to render prompt service by furnishing a list of branch offices or agencies where complete stocks of repair parts are maintained and can be secured within a reasonable time after ordering by part number from the manufacturer's parts book and such discount as may be quoted from year to year by the manufacturer of the vehicle purchased under this specification.

6.7 Statement of Origin or Bill of Sale. A manufacturer's statement of Origin or Bill of Sale showing the applicable purchase order number is required for each vehicle procured under this specification. Unless otherwise specified, such documents shall be forwarded to the consignee.

Military Interest:

Army - TARCOM

Preparing Activity:

GSA-FSS-FYSE

Civil Agency Coordinating Activities:

ERDA
DOT
Interior-BOR-NPS
NASA
U.S. Postal Service

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FOLD

GENERAL SERVICES ADMINISTRATION

POSTAGE AND FEES PAID
U. S.
GENERAL SERVICES ADMINISTRATION

OFFICIAL BUSINESS

National Automotive Center
GSA/FSS/FYSE
CM Bldg. #4, Room 903
Washington, DC 20406

FOLD

GSA FORM FEB 69 2200

[redacted]

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION KKK-T-001985, Interim Federal Specification, Trucks Light Commercial, Electric Powered

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM GSA FEB 69 2200

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